INCH-POUND

MIL-T-82824(OS)
29 December 1989
SUPERSEDING
WS 20882A
1 August 1986

MILITARY SPECIFICATION

TEPANOL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of reaction product OF tetraethylenepentamine acrylonitrile and glycidol referred to herein as tepanol (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Chemical and physical properties.** Chemical and physical properties of the tepanol shall conform to the requirements specified in table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Secondary amine content, equivalent/100 gram (eq/100 g)	0.5	0.7
Total amine content, eq/100 g	0.8	1.2
Tertiary amine content, eq/100 g	0.3	0.5
Hydroxyl content, eq/100 g	0.4	0.6
Ammonia content, ppm	---	500
Moisture, percent	---	0.5
Viscosity, centipoise	30,000	60,000

3.3 **Safety.**

3.3.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.4).

3.4 **Shelf life.** The tepanol shall meet the requirements of this specification for a period of 12 months after the date of quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored at 0 degrees Fahrenheit (°F) maximum.

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3.4.1 **Temperature excursions during storage.** Excursions in temperature environments other than the storage requirements shall be allowed. The cumulative temperature excursion time shall be 336 hours, maximum. Tepanol that has exceeded a cumulative time of 336 hours of temperature excursions above the storage requirement shall be retested to the requirements specified herein and may have the storage life extended as specified in 3.4.2. If the storage temperature exceeds 100°F, for any period of time, the tepanol shall be retested to the requirements specified herein and may have the storage life extended as specified in 3.4.2.

3.4.2 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the tepanol, upon retest, conforms to the hydroxyl content and viscosity requirements. The ammonia content shall be recorded.

NOTE: The requirements specified in 3.4.1, and 3.4.2 are applicable to procuring activity only.

3.5 **Workmanship.** The workmanship shall be such that the tepanol is free from contamination and defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.3.1).

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4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the examination and tests specified in table II. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in Table II.

TABLE II. First article and quality conformance inspections.

Examination and tests	Requirement	Test method
Visual examination	3.5 and section 5	4.5.1
Secondary amine content test	3.2	4.5.2
Total amine content test	3.2	4.5.3
Tertiary amine content test	3.2	4.5.4
Hydroxyl content test	3.2	4.5.5
Ammonia content test	3.2	4.5.6
Moisture test	3.2	4.5.7
Viscosity test	3.2	4.5.8

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes a lot shall consist of all the tepanol subject to inspection at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

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4.4.1.2 **Selection of sample containers.** Sample containers of the tepanol shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.8. Sample material shall be selected as specified in 4.4.1.3.

TABLE III. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1 - 3	All
4 - 40	3
41 - 65	4
66 - 110	5

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.8.

4.4.1.3.1 **Sample preparation.** The selected single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each container shall be examined for conformance to 3.5 and section 5.

4.5.2 **Secondary amine content test.** Secondary amine content shall be determined as follows:

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a. Weigh by difference, 0.25 to 0.35 grams (g) of tepanol to the nearest 0.0001 g. Add 50 ± 10 milliliters (ml) of pyridine, 25 ml of 2-propanol, and 25 ml of distilled water; and stir to dissolve.

b. Add, while stirring, 5 ml of carbon disulfide.

NOTE: Carbon disulfide must be added immediately prior to sample titration.

c. Titrate the sample with standardized 0.25 ± 0.05 N aqueous sodium hydroxide using an automatic titrator (Metrohm Model EA-436 or equivalent, equipped with a glass-Ag/AgCl combination electrode, Metrohm Model EA-120, or equivalent) following the manufacturer's instructions. Record the titration volume to the midpoint of the break as VS. Suggested settings for the Metrohm EA-436:

Selector:	pH/MV
Span:	10 pH
Titration Speed:	2 (1 ml/min)
Automatic Speed Control:	Off
Pen Position:	Far Left

d. Determine at least two reagent blanks, omitting only the sample in a. The blanks must agree to within 0.03 ml. Record the average blank volume as VB.

e. Calculate secondary amine content (eq/100 g) as:

$$\text{Secondary amine, (eq/100 g)} = \frac{(VS - VB) \times N}{10 \times W}$$

Where: VS = sample titration (ml)

VB = blank titration (ml)

N = normality of sodium hydroxide titrant

W = sample weight (g)

10 = factor to convert to percent in this equation

f. Record the secondary amine content to the nearest 0.1 (eq/100 g).

4.5.3 **Total amine content test.** Total amine content shall be determined as follows:

a. Weigh by difference 0.25 to 0.30 g of tepanol to the nearest 0.1 mg. Add 50 ± 10 ml of acetic acid, and stir to dissolve.

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b. Titrate the sample with standardized 0.5N p-toluenesulfonic acid (prepared by dissolving 95 g of p-toluenesulfonic acid in 900 ± 180 ml of glacial acetic acid, carefully adding 20 g acetic anhydride with constant stirring, diluting to 1 liter with glacial acetic acid and standardizing against potassium acid phthalate in glacial acetic acid) using an automatic titrator (Metrohm Model EA-436 or equivalent, equipped with a glass-Ag/AgCl combination electrode, Metrohm Model EA-120, or equivalent) following the manufacturer's instructions. Record the titration volume to the midpoint of the break as V. Suggested settings for the Metrohm EA-436:

Selector:	pH/mV
Span:	5 pH
Titration Speed:	2 (1 ml/min)
Automatic Speed Control:	Off
Pen Position:	Far Left.

c. Calculate p-toluenesulfonic acid normality as:

$$\text{Normality (N)} = \frac{W}{0.20423\ V}$$

Where: W = weight of potassium acid phthalate (g)

V = titration volume of p-toluenesulfonic acid (ml)

0.20423 = gravimetric factor for potassium acid phthalate in this equation

d. Calculate total amine content (eq/100 g) as:

$$\text{Total amine (eq/100 g)} = \frac{VN}{10W}$$

Where: V = titration volume (ml)

N = normality of p-toluenesulfonic acid

W = sample weight (g)

10 = factor to convert to percent in this equation

e. Record the total amine content to the nearest 0.1 eq/100 g.

4.5.4 **Tertiary amine content test.** Tertiary amine content shall be determined from the total amine and secondary amine as follows:

a. Tertiary amine (eq/100 g) = total amine (eq/100 g) - secondary amine (eq/100 g).

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b. Record tertiary amine content to the nearest 0.1 eq/100 g.

4.5.5 **Hydroxyl content test.** Hydroxyl content shall be determined as follows:

a. Weigh a 0.4-g to 0.5-g sample to the nearest 0.1 mg in a dried container which can be tightly closed.

b. Add by pipette 20.0 ml of acetylating reagent (prepared daily by diluting 30 ml of acetic anhydride to 500 ml with pyridine), using a uniform drainage time for the pipette, seal the container, and swirl to dissolve the sample.

NOTE: Erlenmeyer flasks with ground glass stoppers or 8-ounce bottles with caps securely tightened and wrapped with several layers of vinyl tape have been found suitable for this test.

c. Heat the sealed container (the container may be placed in a fabric bag before heating) in a steam bath cavity for 60 to 70 minutes. Make sure the water level in the steam bath covers the liquid level in the container. Cool for a few minutes, and add 50 $\pm$ 5 ml of deionized water.

d. Return to the steam table for 5 minutes, remove from the heat, cool to room temperature and add phenolphthalein solution.

e. Rinse the container closure and walls with a small amount of deionized water, and titrate with standardized 0.5 normal (N) sodium hydroxide to the first permanent pink phenolphthalein endpoint. Record the sample titration volume as VS.

f. Determine a reagent blank in at least triplicate, omitting only the sample and carrying the blanks through the entire procedure. The titration volume of the blanks must agree to within 0.1 ml.

g. Calculate the hydroxyl content as:

$$\text{Hydroxyl content (eq/100 g)} = \frac{(VB-VS) \times N}{10W} - NH$$

Where: VB = Blank titration (ml)

VS = sample titration (ml)

N = normality of the sodium hydroxide

W = sample weight (g)

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NH = secondary amine content eq/100 g (see 4.5.2 f)

10 = factor to convert to eq/100 g in this equation

h. Record hydroxyl content to the nearest 0.1 (eq/100 g).

4.5.6 *Ammonia content test.* Ammonia content shall be determined as follows:

a. Prepare a minimum of two ammonia standards in the range from 10 to 100 ppm. To prepare a 100 ppm as anhydrous ammonia standard, dissolve 0.6288 g of ammonium chloride in 2000 ml of distilled water. One to one dilution of this standard solution yields 50 ppm as anhydrous ammonia. Nine to one dilution yields 10 ppm as anhydrous ammonia.

b. Assemble an ammonia electrode using the manufacturer's instructions, and calibrate using 100 ± 1 ml of the standard plus 1 ml of 10N NaOH and stirring for 1 minute.

NOTE: A millivolt meter or a direct reading instrument may be used. A plot of millivolts versus ppm anhydrous ammonia is required for extrapolation/interpolation of millivolt data.

NOTE: Some magnetic stirrers generate enough heat to affect the electrode's response. An insulating layer, such as cork or plastic foam, between the stirrer and beaker may be required.

c. Weigh 2 to 3 grams of sample to the nearest 0.1 mg in a beaker, and add 100 ± 1 ml of distilled water and 1 ml of 10N NaOH with stirring.

d. Calculate the ammonia concentration in the sample solution by either (1) or (2) as follows:

(1) Record the ammonia concentration in the solution to the nearest 10 ppm from a direct reading instrument, and calculate the ammonia content as follows:

$$\text{Ammonia content, anhydrous ammonia ppm} = \frac{100C}{W}$$

Where: C = sample solution concentration (from standard curve)

W = sample weight, g

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(2) Plot the millivolt reading versus the ppm in the ammonia standards, and determine the ammonia ppm in the sample solution from the plot. Calculate the ammonia content as follows:

$$\text{Ammonia content, anhydrous ammonia ppm} = \frac{100C}{W}$$

Where: C = sample solution concentration (from standard curve)

W = sample weight, g

e. Record ammonia content to the nearest 10 ppm.

4.5.7 **Moisture test.** Moisture shall be determined as follows:

a. Weigh a 3-g to 4-g sample to the nearest 0.01 g.

b. Dissolve the sample in 100 ± 20 ml of 1 to 1 pyridinemethanol carrier solution which has been neutralized with Karl Fischer (KF) reagent.

c. Titrate the moisture using an automatic titrator with standardized dilute KF reagent (prepared by diluting commercial KF reagent with pyridine or KF reagent diluent to a water equivalent of 1.8 to 2.3 mg of water per ml of KF reagent).

d. Consider the determination complete when an endpoint is reached which is stable for 30 seconds.

e. Calculate the total moisture as follows:

$$\text{Percent moisture} = \frac{V \times E}{10W}$$

Where: V = volume of standardized KF reagent, ml

E = water equivalent of standardized KF reagent, mg/ml

W = weight of sample, g.

10 = unit/percentage conversion factor

f. Record total moisture to the nearest 0.01 percent.

4.5.8 **Viscosity test.** Viscosity shall be determined as specified in 4.5.8.1 or 4.5.8.2.

4.5.8.1 **Brookfield.** Brookfield viscosity shall be determined as follows:

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a. Condition the sample in a constant temperature bath set at 25.0 ± 0.1° Celsius (C).

b. Set the speed of a Brookfield Model RVF viscometer at 4 rpm. Attach a No. 5 spindle to the viscometer, and support the viscometer so as to center the spindle in a pint-sized sample container. Adjust the viscometer until the spindle index mark is even with the surface of the sample.

c. Turn on the viscometer, and take five readings at 1-minute intervals.

d. Calculate the Brookfield viscosity as follows: $V = f \times R$

Where: V = viscosity of sample (centipoise) at 25°C

f = factor of viscometer

R = scale reading at 25°C.

e. Record Brookfield viscosity at 25°C to the nearest 1000 centipoise.

4.5.8.2 **Cannon-Manning.** Cannon-Manning viscosity shall be determined as follows:

a. Fill a calibrated Cannon-Manning vacuum viscometer tube to within ± 0.5 inch of the "Full" line, allowing the sample to run down side of tube to prevent air entrapment.

b. Condition the viscometer with the sample in a constant temperature bath at 25.0 ± 0.1°C for a minimum of 30 minutes.

c. Exclude all air from the sample before starting the run.

d. Using a vacuum control capable of controlling to ± 0.1 centimeter, stabilize the regulator at 30 centimeters of mercury and attach the regulator to viscometer.

e. Read the efflux time for the small bulb and for the large bulb.

f. Calculate the viscosity for each bulb as follows:

Viscosity, poise = T x C

Where: T = efflux time, seconds

C = calibration factor for bulb

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g. Average the results of the calculations.

h. Record the average to the nearest 1000 centipoise.

4.6 **Inspection of packaging.** The tepanol containers and container markings shall be visually examined to verify compliance with section 5.

5. **PACKAGING**

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The tepanol shall be packaged to afford adequate protection against loss and contamination. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178 or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTH PERIOD PROVIDED THE TEPANOL, UPON RETEST, CONFORMS TO THE HYDROXYL CONTENT AND VISCOSITY REQUIREMENTS OF MIL-T-82824".

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tepanol specified herein is intended for use as a bonding agent in the Standard Missile MK-104 Dual Thrust Rocket Motor propellant manufacture.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheet (see 3.3.2 and 6.4).

e. Packaging required if other than as specified.

f. Special marking if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the tepanol sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheet (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MDSDs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.5 **Definitions.**

6.5.1 **Batch.** A batch is as that amount of tepanol made as one unit in the manufacturing process.

6.5.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the tepanol is accumulated for a lot.

6.6 **Subject term (key word) listing.**

Bonding agent
Rocket Motor, Dual Thrust, Mark 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1376-N365)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82824	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
TEPANOL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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